

Lecture 5 : Limits

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1. Right hand limit

Let $D \subseteq \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$ is a function. Let c be a limit point of $D_1 = D \cap (c, \infty) = \{x \in D : x > c\}$. Function f is said to have a **right hand limit** $l (\in \mathbb{R})$ at c if corresponding to a preassigned positive number ϵ , there exists a positive number δ , such that

$$\begin{aligned} & |f(x) - l| < \epsilon \text{ for all } x \in N'(c, \delta) \cap D_1 \\ \text{i.e. } & l - \epsilon < f(x) < l + \epsilon \text{ for all } x \in D \text{ satisfying } c < x < c + \delta. \end{aligned}$$

In this case we write $\lim_{x \rightarrow c^+} f(x) = l$.

2. Left hand limit

Let $D \subseteq \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$ is a function. Let c be a limit point of $D_2 = D \cap (-\infty, c) = \{x \in D : x < c\}$. Function f is said to have a **left hand limit** $l (\in \mathbb{R})$ at c if corresponding to a preassigned positive number ϵ , there exists a positive number δ , such that

$$\begin{aligned} & |f(x) - l| < \epsilon \text{ for all } x \in N'(c, \delta) \cap D_2 \\ \text{i.e. } & l - \epsilon < f(x) < l + \epsilon \text{ for all } x \in D \text{ satisfying } c - \delta < x < c. \end{aligned}$$

In this case we write $\lim_{x \rightarrow c^-} f(x) = l$.

Theorem 1 (Sandwich Theorem) Let $D \subseteq \mathbb{R}$ and f, g, h are functions on $D \rightarrow \mathbb{R}$. Let $c \in D'$. If $f(x) \leq g(x) \leq h(x)$ for all $x \in D - \{c\}$ and $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} h(x) = l$, then $\lim_{x \rightarrow c} g(x) = l$.

(For proof, see the reference book.)

Theorem 2 Let $D \subseteq \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$ be a function. Let c be a limit point of both the sets $D_1 (= D \cap (c, \infty))$ and $D_2 (= D \cap (-\infty, c))$. Then, $\lim_{x \rightarrow c} f(x) = l \in \mathbb{R}$, if and only if $\lim_{x \rightarrow c^+} f(x) = l = \lim_{x \rightarrow c^-} f(x)$.

(For proof, see the reference book.)

Examples

1. Evaluate the one sided limits - (i) $\lim_{x \rightarrow 0^-} [x]$, (ii) $\lim_{x \rightarrow 0^+} [x]$ and show that, $\lim_{x \rightarrow 0} [x]$ does not exist (here, $[x]$ is greatest integer of x).

Solution : Let $f(x) = [x]$. The domain of f is $\mathbb{R}$.

Since, we have to evaluate $\lim_{x \rightarrow 0^-} f(x)$ and $\lim_{x \rightarrow 0^+} f(x)$, thus, we are interested in the nature of the function f in the neighborhood of 0. We know that

$$f(x) = \begin{cases} -1, & \text{if } -1 < x < 0 \\ 0, & \text{if } 0 \leq x < 1. \end{cases}$$

Therefore, $\lim_{x \rightarrow 0^-} f(x) = -1$ and $\lim_{x \rightarrow 0^+} f(x) = 0$.

Since, $\lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x)$ and therefore, $\lim_{x \rightarrow 0} f(x)$ does not exist.

2. Evaluate $\lim_{x \rightarrow 3} \left([x] - \left[\frac{x}{3} \right] \right)$.

Solution : Let $f(x) = \left([x] - \left[\frac{x}{3} \right] \right)$. The domain of f is $\mathbb{R}$.

Since, we have to evaluate $\lim_{x \rightarrow 3} f(x)$, thus, we are interested in the nature of the function f in the neighborhood of 3. We know that

$$[x] = \begin{cases} 2, & \text{if } 2 < x < 3 \\ 3, & \text{if } 3 \leq x < 4. \end{cases}$$

and

$$\left[\frac{x}{3} \right] = \begin{cases} 0, & \text{if } 0 < x < 3 \\ 1, & \text{if } 3 \leq x < 6. \end{cases}$$

Therefore,

$$f(x) = \begin{cases} 2, & \text{if } 2 < x < 3 \\ 2, & \text{if } 3 \leq x < 4. \end{cases}$$

Since, $\lim_{x \rightarrow 3^-} f(x) = 2$ and $\lim_{x \rightarrow 3^+} f(x) = 2$. Therefore, $\lim_{x \rightarrow 3} f(x) = 2$

3. Prove that, $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.

Solution : In $0 < x < \frac{\pi}{2}$, $\sin x < x < \tan x$.

Therefore, $1 < \frac{x}{\sin x} < \frac{1}{\cos x}$ in $0 < x < \frac{\pi}{2}$.

$\therefore \lim_{x \rightarrow 0^+} \frac{1}{\cos x} = 1$, therefore, by Sandwich theorem we have, $\lim_{x \rightarrow 0^+} \frac{x}{\sin x} = 1$.

Thus, $\lim_{x \rightarrow 0^+} \frac{\sin x}{x} = 1$.

In $-\frac{\pi}{2} < x < 0$, $\tan x < x < \sin x$.

Therefore, $1 < \frac{x}{\sin x} < \frac{1}{\cos x}$ in $-\frac{\pi}{2} < x < 0$.

$\therefore \lim_{x \rightarrow 0^-} \frac{1}{\cos x} = 1$, therefore, by Sandwich theorem we have, $\lim_{x \rightarrow 0^-} \frac{x}{\sin x} = 1$.

Thus, $\lim_{x \rightarrow 0^-} \frac{\sin x}{x} = 1$.

Since, $\lim_{x \rightarrow 0^+} \frac{\sin x}{x} = \lim_{x \rightarrow 0^-} \frac{\sin x}{x} = 1$. It follows that, $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.

3. Infinite limits

1. Let $D \subseteq \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$ be a function. Let c be a limit point of D . If corresponding to a preassigned positive number G , there exists a positive number δ , such that

$$f(x) > G \text{ for all } x \in N'(c, \delta) \cap D,$$

then we say that, f tends to ∞ as $x \rightarrow c$, and we write $\lim_{x \rightarrow c} f(x) = \infty$.

2. Let $D \subseteq \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$ be a function. Let c be a limit point of D . If corresponding to a preassigned positive number G , there exists a positive number δ , such that

$$f(x) < -G \text{ for all } x \in N'(c, \delta) \cap D,$$

then we say that, f tends to $-\infty$ as $x \rightarrow c$, and we write $\lim_{x \rightarrow c} f(x) = -\infty$.

Example : Show that $\lim_{x \rightarrow 0} f(x) = \infty$, where $f(x) = \frac{1}{x^2}$.

Solution : In every neighborhood of 0, f is unbounded above.

Let us choose $G > 0$. Then $f(x) > G$ for all x satisfying $-\frac{1}{\sqrt{G}} < x < \frac{1}{\sqrt{G}}$, $x \neq 0$.

This implies, $f(x) > G$ for all $x \in N'(0, \delta)$ where $\delta = \frac{1}{\sqrt{G}}$.

Therefore, $\lim_{x \rightarrow 0} f(x) = \infty$.

4. Limit at infinity

1. Let $D \subseteq \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$ be a function. Let $(c, \infty) \subset D$ for some $c \in \mathbb{R}$. We say that, $f \rightarrow l$ ($l \in \mathbb{R}$) as $x \rightarrow \infty$, if corresponding to a preassigned positive number ϵ , there exists a real number $G > c$, such that

$$|f(x) - l| < \epsilon \text{ for all } x > G.$$

In this case, we write $\lim_{x \rightarrow \infty} f(x) = l$.

2. Let $D \subseteq \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$ be a function. Let $(-\infty, c) \subset D$ for some $c \in \mathbb{R}$. We say that, $f \rightarrow l (l \in \mathbb{R})$ as $x \rightarrow -\infty$, if corresponding to a preassigned positive number ϵ , there exists a real number $G < c$, such that

$$|f(x) - l| < \epsilon \text{ for all } x < G.$$

In this case, we write $\lim_{x \rightarrow -\infty} f(x) = l$.

Example : Show that $\lim_{x \rightarrow \infty} f(x) = 0$, where $f(x) = \frac{1}{x}$.

Solution : The domain of f is $D = \mathbb{R} - \{0\}$. Thus, $(0, \infty) \subset D$.

Let us choose $\epsilon > 0$.

When $x \neq 0$, $|f(x) - 0| = \frac{1}{x}$.

Therefore, $|f(x) - 0| < \epsilon$ for all $x > G$, where $G = 1/\epsilon > 0$.

Hence, $\lim_{x \rightarrow \infty} f(x) = 0$

Reference book : *Introduction to real analysis by S.K. Mapa.*